16. Which one of the following orders is correct for the bond dissociation enthalpy of halogen molecules? (NEET)

a) $Br_2 > I_2 > F_2 > Cl_2$ b) $F_2 > Cl_2 > Br_2 > I_2$

c) $I_2 > Br_2 > Cl_2 > F_2$ d) $Cl_2 > Br_2 > F_2 > I_2$

17. Among the following the correct order of acidity is (NEET)

a) $HClO_2 < HClO < HClO_3 < HClO_4$ b) $HClO_4 < HClO_2 < HClO < HClO_3$

c) $HClO_3 < HClO_4 < HClO_2 < HClO$ d) $HClO < HClO_2 < HClO_3 < HClO_4$

18. When copper is heated with conc HNO_3 it produces

a) $Cu(NO_3)_2$, NO and NO_2 b) $Cu(NO_3)_2$ and N_2O

c) $Cu(NO_3)_2$ and NO_2 d) $Cu(NO_3)_2$ and NO

Answer the following questions:

1. What is inert pair effect?
2. Chalcogens belongs to p-block. Give reason.
3. Explain why fluorine always exhibit an oxidation state of -1?
4. Give the oxidation state of halogen in the following.

a) OF_2 b) O_2F_2 c) Cl_2O_3 d) I_2O_4

5. What are interhalogen compounds? Give examples.
6. Why fluorine is more reactive than other halogens?
7. Give the uses of helium.
8. What is the hybridisation of iodine in IF_7? Give its structure.
9. Give the balanced equation for the reaction between chlorine with cold NaOH and hot NaOH.
10. How will you prepare chlorine in the laboratory?
11. Give the uses of sulphuric acid.
12. Give a reason to support that sulphuric acid is a dehydrating agent.
13. Write the reason for the anamolous behaviour of Nitrogen.
14. Write the molecular formula and structural formula for the following molecules.

a) Nitric acid b) dinitrogen pentoxide

c) phosphoric acid d) phosphine

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